CAPITAL MARKET THEORY: CML - LEVERAGED PORTFOLIOS

Learning Outcome

- explain the capital allocation line (CAL) and the capital market line (CML)

In the previous example, Mr. Miles evaluated an investment of between 0 percent and 100 percent in the market and the balance in T-bills. The line connecting R_f and M (market portfolio) in Exhibit 4 illustrates these portfolios with their respective levels of investment. At R_f, an investor is investing all of his or her wealth into risk-free securities, which is equivalent to lending 100 percent at the risk-free rate. At Point M he or she is holding the market portfolio and not lending any money at the risk-free rate. The combinations of the risk-free asset and the market portfolio, which may be achieved by the points between these two limits, are termed "lending" portfolios. In effect, the investor is lending part of his or her wealth at the risk-free rate.

If Mr. Miles is willing to take more risk, he may be able to move to the right of the market portfolio (Point M in Exhibit 4) by borrowing money and purchasing more of Portfolio M. Assume that he is able to borrow money at the same risk-free rate of interest, R_f, at which he can invest. He can then supplement his available wealth with borrowed money and construct a borrowing portfolio. If the straight line joining R_f and M is extended to the right of Point M, this extended section of the line represents borrowing portfolios. As one moves further to the right of Point M, an increasing amount of borrowed money is being invested in the market. This means that there is *negative* investment in the risk-free asset, which is referred to as a *leveraged position* in the risky portfolio. The particular point chosen on the CML will depend on the individual's utility function, which, in turn, will be determined by his risk and return preferences.

EXAMPLE 2

Risk and Return of a Leveraged Portfolio with Equal Lending and Borrowing Rates

1. Mr. Miles decides to set aside a small part of his wealth for investment in a portfolio that has greater risk than his previous investments because he anticipates that the overall market will generate attractive returns in the future. He assumes that he can borrow money at 5 percent and achieve the same return on the S&P 500 as before: an expected return of 15 percent with a standard deviation of 20 percent.

   Calculate his expected risk and return if he borrows 25 percent, 50 percent, and 100 percent of his initial investment amount.

   Hide Solution

   Solution:

   The leveraged portfolio's standard deviation and return can be calculated in the same manner as before with the following equations:

   $$E(R_p) = w_1R_f + (1 - w_1)E(R_m)$$

   and

   $$\sigma_p = (1 - w_1)\sigma_m$$

Rate Your Confidence
High
Medium
Low
Continue »
Category
Portfolio Risk and Return: Part II
Related Questions:
Practice questions related to this topic

is borrowed, $w_1 = -0.25$, and $(1 - w_1) = 1.25$, etc.

Return with $w_1 = -0.25 = (-0.25 \times 5\%) + (1.25 \times 15\%) = 17.5\%$.

Standard deviation with $w_1 = -0.25 = 1.25 \times 20\% = 25\%$.

Return with $w_1 = -0.50 = (-0.50 \times 5\%) + (1.50 \times 15\%) = 20.0\%$.

Standard deviation with $w_1 = -0.50 = 1.50 \times 20\% = 30\%$.

Return with $w_1 = -1.00 = (-1.00 \times 5\%) + (2.00 \times 15\%) = 25.0\%$.

Standard deviation with $w_1 = -1.00 = 2.00 \times 20\% = 40\%$.

Note that negative investment (borrowing) in the risk-free asset provides a higher expected return for the portfolio but that higher return is also associated with higher risk.

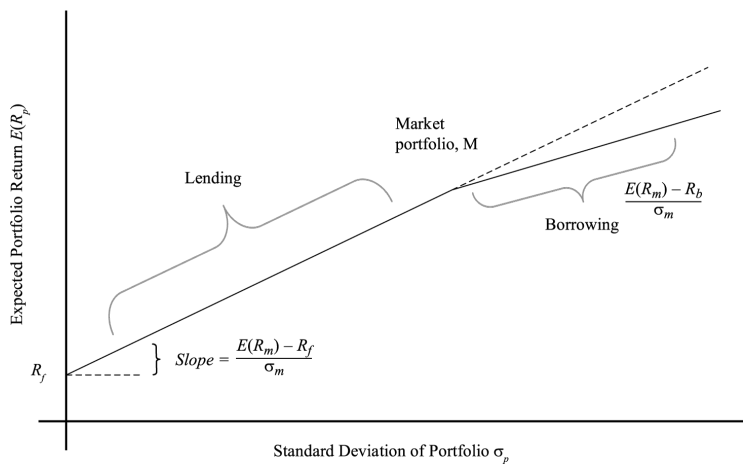
Leveraged Portfolios with Different Lending and Borrowing Rates

Although we assumed that Mr. Miles can borrow at the same rate as the US government, it is more likely that he will have to pay a higher interest rate than the government because his ability to repay is not as certain as that of the government. Now consider that although Mr. Miles can invest (lend) at R_f, he can borrow at only R_b, a rate that is higher than the risk-free rate.

With different lending and borrowing rates, the CML will no longer be a single straight line. The line will have a slope of $[E(R_m) - R_f]/\sigma_m$ between Points R_f and M, where the lending rate is R_f, but will have a smaller slope of $[E(R_m) - R_b]/\sigma_m$ at points to the right of M, where the borrowing rate is R_b. Exhibit 5 illustrates the CML with different lending and borrowing rates.

Exhibit 5:

CML with Different Lending and Borrowing Rates



▶ Expand / collapse Extended Description

The equations for the two lines are given below.

$$w_1 \geq 0: E(R_p) = R_f + \left(\frac{E(R_m) - R_f}{\sigma_m}\right) \times \sigma_p$$

and

$$w_1 < 0: E(R_p) = R_b + \left(\frac{E(R_m) - R_b}{\sigma_m}\right) \times \sigma_p$$

where borrowing, or negative investment in the risk-free asset, occurs. Note that the only difference between the two equations is in the interest rates used for borrowing and lending.

All passive portfolios will lie on the kinked CML, although the investment in the risk-free asset may be positive (lending), zero (no lending or borrowing), or negative (borrowing). Leverage allows less risk-averse investors to increase the amount of risk they take by borrowing money and investing more than 100 percent in the passive portfolio.

EXAMPLE 3

Leveraged Portfolio with Different Lending and Borrowing Rates

1. Mr. Miles approaches his broker to borrow money against securities held in his portfolio. Even though Mr. Miles' loan will be secured by the securities in his portfolio, the broker's rate for lending to customers is 7 percent. Assuming a risk-free rate of 5 percent and a market return of 15 percent with a standard deviation of 20 percent, estimate Mr. Miles' expected return and risk if he invests 25 percent and 75 percent in the market and if he decides to borrow 25 percent and 75 percent of his initial investment and invest the money in the market.

Hide Solution

Solution:

The unleveraged portfolio's standard deviation and return are calculated using the same equations as before:

$$E(R_p) = w_1R_f + (1 - w_1)E(R_m),$$

and

$$\sigma_p = (1 - w_1)\sigma_m$$

The results are unchanged. The slope of the line for the unleveraged portfolio is 0.50, just as before:

Return with 25 percent invested in the market = (0.75 × 5%) + (0.25 × 15%) = 7.5%.

Standard deviation with 25 percent invested in the market = 0.25 × 20% = 5%.

Return with 75 percent invested in the market = (0.25 × 5%) + (0.75 × 15%) = 12.5%.

Standard deviation with 75 percent invested in the market = 0.75 × 20% = 15%.

For the leveraged portfolio, everything remains the same except that R_f is replaced with R_b.

$$E(R_p) = w_1R_b + (1 - w_1)E(R_m),$$

and

$$\sigma_p = (1 - w_1)\sigma_m.$$

Return with w_1 = –0.25 = (–0.25 × 7%) + (1.25 × 15%) = 17.0%.

Standard deviation with w_1 = –0.25 = 1.25 × 20% = 25%.

Return with w_1 = –0.75 = (–0.75 × 7%) + (1.75 × 15%) = 21.0%.

Standard deviation with w_1 = –0.75 = 1.75 × 20% = 35%.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

market, the expected return increases but so does the standard deviation of the portfolio. The slope of the line for the leveraged portfolio is 0.40, compared with 0.50 for the unleveraged portfolio, which means that for every 1 percent increase in risk, the investor gets a 0.40 percent increase in expected return in the leveraged part of the portfolio, compared with a 0.50 percent increase in expected return in the unleveraged part of the portfolio. Only investors who are less risk averse will choose leveraged portfolios.

Discuss Filter

Discussion

What do you want to discuss... (required)

No messages to display

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search